INCH-POUND
MIL-PRF-19500/198E
25 March 2008
SUPERSEDING
MIL-PRF-19500/198D
4 September 2007

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, THYRISTORS, TYPES 2N1870A, 2N1871A, 2N1872A, AND 2N1874A, JAN

Inactive for new design after 4 September 2007.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon thyristors. One level of product assurance is provided for each device type as specified in MIL-PRF-19500.

* 1.2 Physical dimensions. See figure 1 (TO-5).

1.3 Maximum ratings. (1)

I_O T_A = +100°C	i_f (surge)	V_G V_{CG}	I_G	i_G	Operating and storage temperature	Operating altitude
mA dc	a	V dc	mA dc	ma	°C	mm Hg
220	15	5	25	250	-65 to +150	15

Type	V_{RM} (2)	VFBXM (2)
	V(pk)	V(pk)
2N1870A	30	30
2N1871A	60	60
2N1872A	100	100
2N1874A	200	200

(1) For additional information on ratings see figures 2 and 3.
(2) Blocking voltage ratings apply with gate connected through a resistor, R_2 = 1 kΩ or less, to cathode as shown on figure 4.

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1.4 Primary electrical characteristics.

Limits	V_f i_f = 2A	I_H	V_{GT} V_{AC} = 5 V R_e = 1 kΩ	I_{GT} V_{AC} = 5 V R_e > 1 kΩ
	V	mA dc	V dc	μA dc
Min		0.3	0.4	
Max	2.5	5.0	0.8	200

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

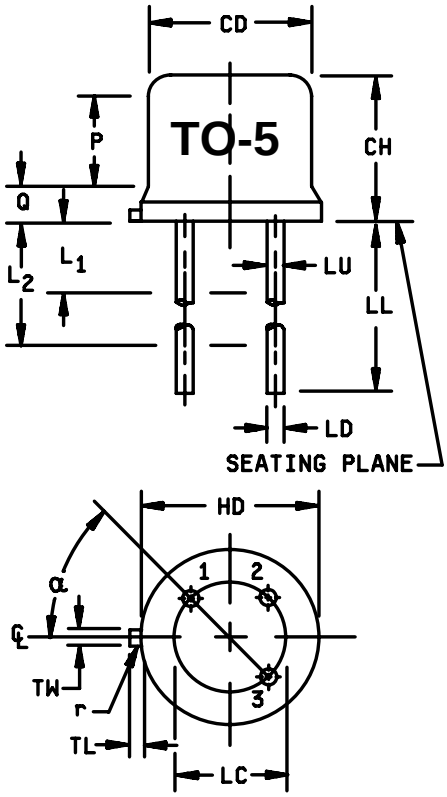
3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML) before contract award (see 4.2 and 6.3).

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.200	.260	5.08	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		
LD	.016	.019	0.41	0.48	3, 7
* LL	1.500	1.750	38.1	44.45	3
LU	.016	.019	0.41	0.48	7
L_1		.050		1.27	7
L_2	.250		6.35		7
P	.100		2.54		6
Q		.030		0.76	8, 10
TL	.029	.045	0.74	1.14	11
TW	.028	.034	0.71	0.86	11
r		.010		0.25	11, 12
α	45° TP		45° TP		



NOTES:

1. Dimension are in inches.
2. Millimeters are given for general information only.
3. All terminals are electrically insulated from the case.
4. Leads may be soldered to within .063 inch (1.60 mm) of base. Heat sinking not required if temperature-time exposure is less than 230°C for 10 seconds.
5. This device is for socketed single-sided circuit-board, wire-in, and similar applications. Where used in double sided or eyeletted circuit-board, or similar applications where solder bridging may occur, a dielectric washer or other standoff device may be necessary.
6. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed .010 inch (0.25 mm).
7. The specified lead diameter applies in the zone between .050 (1.27 mm) and .250 inch (6.35 mm) from the base seat. Between .250 inch (6.35 mm) and 1.5 (38.1 mm) a maximum of .021 inch (0.533 mm) diameter is held. Outside of these zones the lead diameter is not controlled.
8. Body contour optional within zone defined by Q.
9. Lead 1 = cathode, lead 2 = gate, lead 3 = anode.
10. Details of outline in this zone are optional.
11. Tab is optional.
12. Dimension r (radius) applies to both inside corners of tab.
13. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

* FIGURE 1. Physical dimensions of semiconductor types 2N1870A, 2N1871A, 2N1872A and 2N1874A (TO-5).

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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500, and as follows:

I_{FBX}	DC forward blocking current with resistor R_2 across gate and cathode.
I_{RBX}	DC reverse blocking current with resistor R_2 across gate and cathode.
i_G	Peak gate current.
V_{AA}	Anode power supply voltage (dc).
V_{CG}	Reverse gate voltage.
V_{FBX}	Forward anode voltage.
V_G	Rated reverse gate voltage.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, table I herein, and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.3.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIb (JAN) of MIL-PRF-19500 and 4.3.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2.

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4.3.2.1 Group B inspection, table VIb (JAN) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1040	Operational life, see 4.6, test circuit on figure 6, T_A = +100°C.
B6	1031	T_A = +150°C.

4.3.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Lead fatigue.
C6	1031	T_A = +150°C.

4.4 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.4.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5 Holding current. The holding current is measured by decreasing the anode current to the point where the device under test switches from the on-state to the off-state. The gate current shall be -150 μA and the anode supply voltage shall be 5 volts. The test circuit shown on figure 5, or its equivalent, may be used. Resistor R_4 is increased, thereby decreasing device anode current, until the device switches to the off-state. The current measured by M_1 immediately prior to switching is the holding current. Switch S_1 is used to trigger the device to the on-state prior to making a holding current measurement.

4.6 Operational life test. The operational life test shall be performed in the test circuit shown on figure 6, or its equivalent. Transformer T_1 is adjusted such that the peak value of secondary voltage from T_3 equals the rated forward blocking voltage of the device under test. Transformer T_2 is adjusted such that meter reads 220 mA average rectified current. Transformer T_5 is selected to provide a maximum secondary peak voltage of 5 volts as read on meter M3. The device under test shall be in an oven at an ambient temperature of +100° ±5°C.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward blocking current	4206	DC method, bias condition B; R_2 = 1,000 Ω, .25 watt	I_{FBX}		10	μA dc
2N1870A		V_{FBX} = 30				
2N1871A		V_{FBX} = 60				
2N1872A		V_{FBX} = 100				
2N1874A		V_{FBX} = 200				
Reverse blocking current	4211	DC method, bias condition B; R_2 = 1,000 Ω, .25 watt	I_{RBX}		10	μA dc
2N1870A		V_{FBX} = 30				
2N1871A		V_{FBX} = 60				
2N1872A		V_{FBX} = 100				
2N1874A		V_{FBX} = 200				
Gate trigger current	4221	$V_2 = V_{FBX}$ = 5 V dc R_e = 1,000 Ω; R_L = 100 Ω;	I_{GT}		200	μA dc
Gate trigger voltage	4221	$V_2 = V_{FBX}$ = 5 V dc R_e = 1,000 Ω R_L = 100 Ω;	V_{GT}	0.4	0.8	V dc
Forward “on” voltage	4226	I_{FM} = 2 a(pk) (pulse); pulse width = 8.5 ms; maximum; duty cycle = 2 percent maximum	V_f		2.5	V dc
Exponential rate of voltage rise	4231	Bias condition B; repetition rate = 60 pps; 50 Ω ≤ R_L ≤ 400 Ω; C = .001 μF min; test duration = 15 sec; R_3 = 1,000 Ω	V_{FBX}			
2N1870A		dv/dt = 100 V/μs; V_{AA} = 30 V dc		25		
2N1871A		dv/dt = 60 V/μs; V_{AA} = 60 V dc		55		
2N1872A		dv/dt = 40 V/μs; V_{AA} = 100 V dc		95		
2N1874A		dv/dt = 30 V/μs; V_{AA} = 200 V dc		190		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Reverse gate current	4219	V_{CG} = 5 V dc, anode open	I_{KG}		10	μA dc
Holding current	4201	I_G = -150 μA dc, V_{AA} = 5 V dc	I_H	0.3	5	mA dc
Subgroup 3						
High temperature operation:	4206	T_A = +125°C				
Forward blocking current	4206	R_2 = 1,000 ohms	I_{FBX}		100	μA dc
2N1870A		V_{FBX} = 30				
2N1871A		V_{FBX} = 60				
2N1872A		V_{FBX} = 100				
2N1874A		V_{FBX} = 200				
Reverse blocking current	4211	R_2 = 1,000 ohms	I_{RBX}		100	μA dc
2N1870A		V_{FBX} = 30				
2N1871A		V_{FBX} = 60				
2N1872A		V_{FBX} = 100				
2N1874A		V_{FBX} = 200				
Gate trigger voltage	4221	$V_2 = V_{FBX}$ = 5 V, R_e = 1,000 Ω R_L = 100 Ω	V_{GT}	0.2		V dc
Holding current	4201	I_G = -150 μA dc, V_{AA} = 5 V dc	I_H	0.2		mA dc
Low temperature operation:		T_A = -65°C				
Gate trigger voltage	4221	$V_2 = V_{FBX}$ = 5 V R_e = 1,000 Ω R_L = 100 Ω	V_{GT}		1.0	V dc
Gate trigger current	4221	$V_2 = V_{FBX}$ = 5 V, R_e = 1,000 Ω R_L = 100 Ω	I_{GT}		500	μA dc
Holding current	4201	I_G = -150 μA dc, V_{AA} = 5 V dc	I_H		15	mA dc

See footnote at end of table.

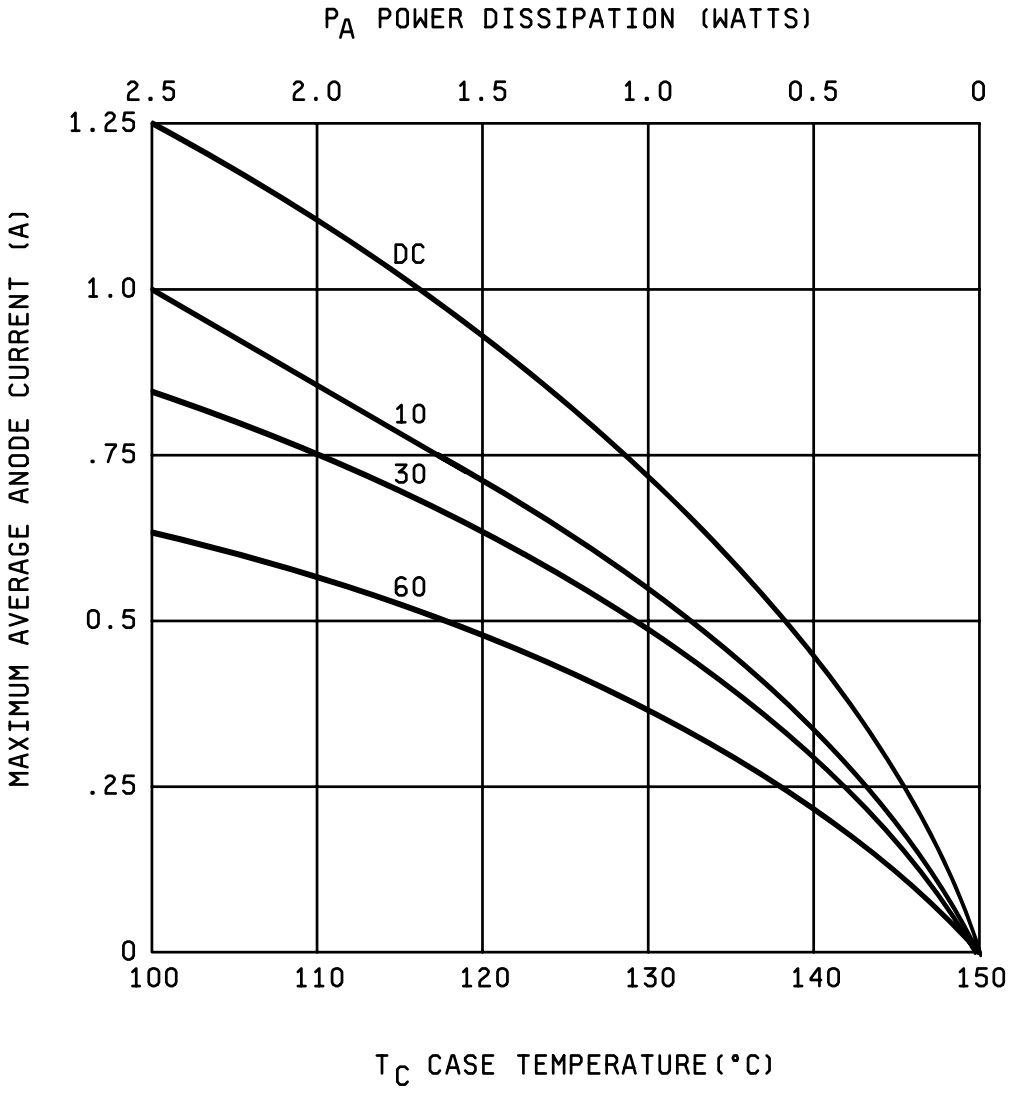
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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroups 4 and 5						
Not applicable						
Subgroup 6						
Surge current	4066	I_F = 15 A(pk) (.5 sine wave); Ten surges per minute; I_O = 250 mA dc at rated V_{RRM}; T_C = +65°C; f = 60 Hz; surge duration = 7 ms, minimum.				
Electrical measurements		See table I, group A, subgroup 2				

1/ For sampling plan, see MIL-PRF-19500.

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NOTE: Ratings apply for 50 to 400 Hz frequency.

FIGURE 2. Anode current rating as a function of operating case temperature.

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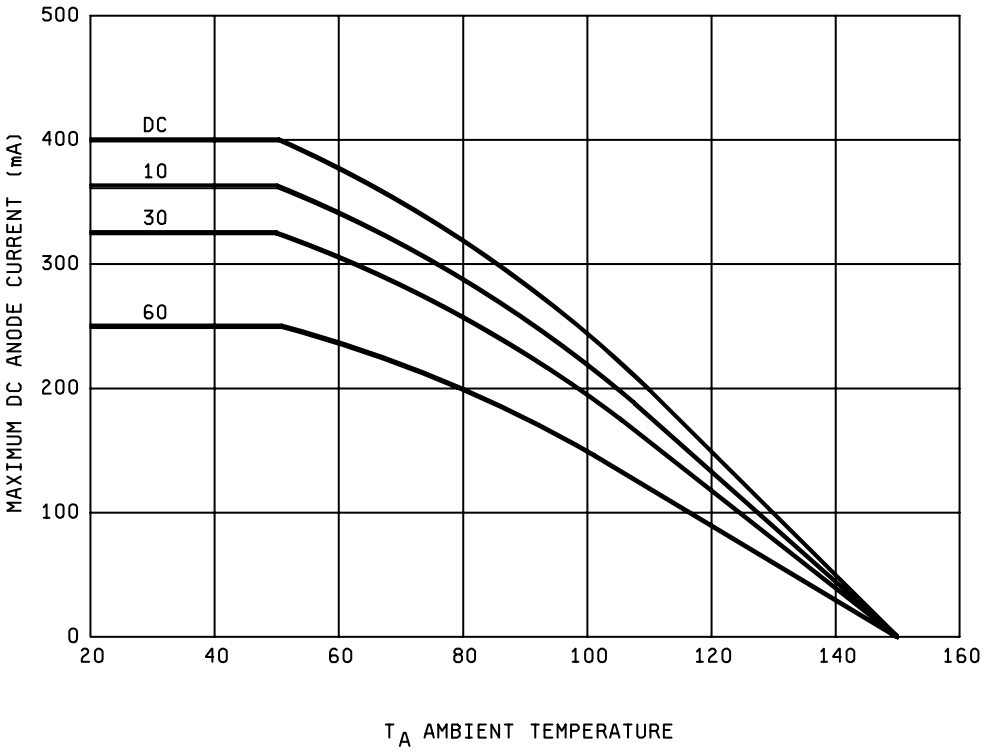


FIGURE 3. Anode current rating as a function of operating ambient temperature.

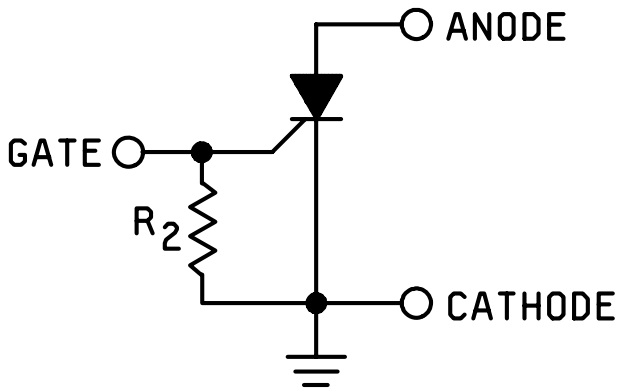
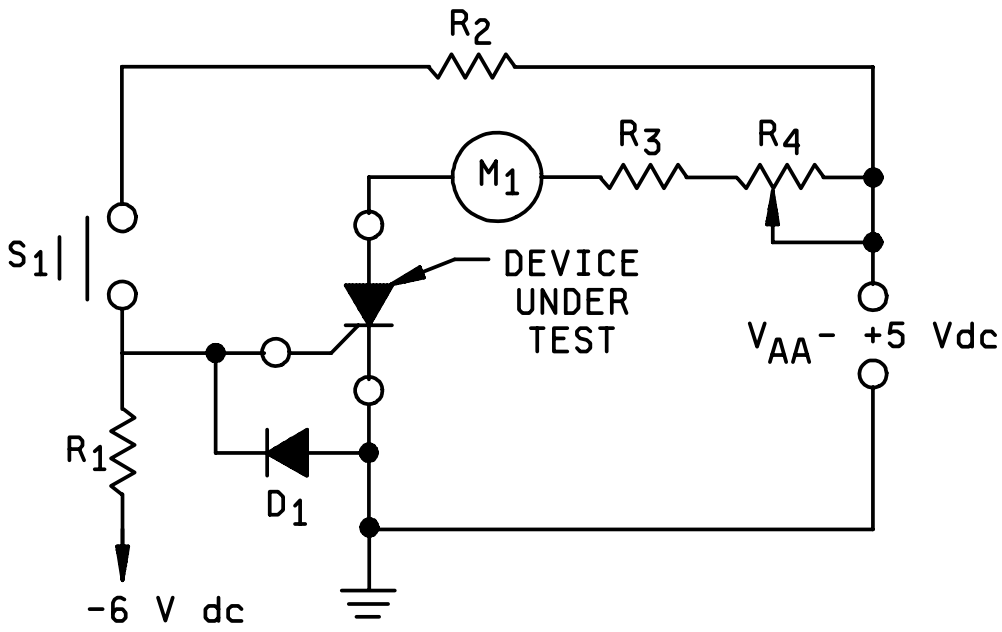


FIGURE 4. Condition for blocking voltage rating.

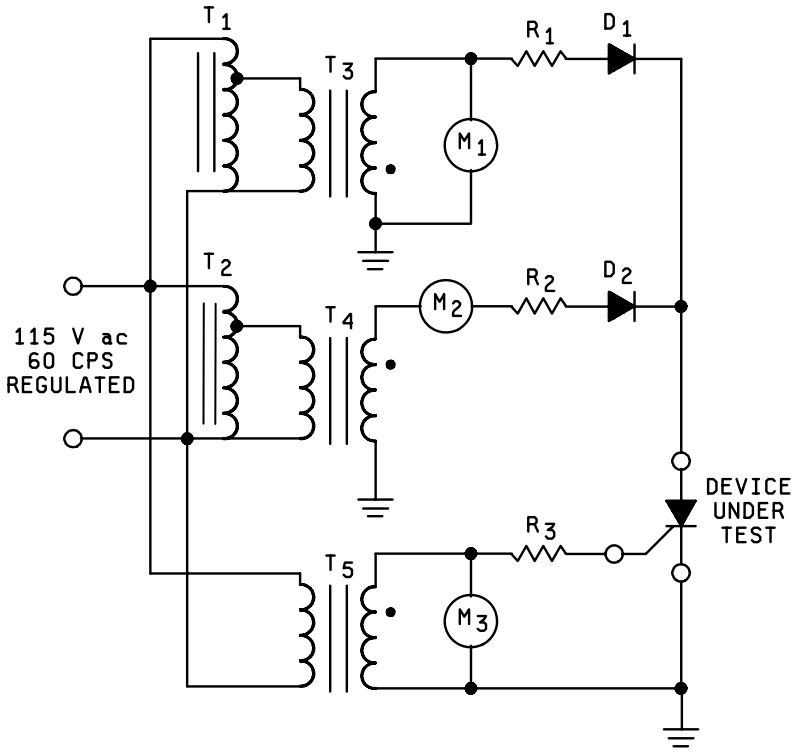
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M_1	DC milliammeter	R_1	40 kΩ, 1/4 watt
S_1	Trigger on switch, momentary type, normally open	R_2	4,700 Ω, 1/4 watt
		R_3	270 Ω, 1/4 watt
D_1	1N457 diode	R_4	25 kΩ potentiometer

FIGURE 5. Test circuit for holding current.

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T1	115 VAC adjustable transformer	R2	20 Ω, 20 watt
T2	115 VAC adjustable transformer	R3	1,000 Ω, 1/4 watt
T3	Step up transformer 115 VAC to 140 VAC	M1	Peak reading voltmeter or RMS equivalent (0 - 250 volts)
T4	Step down transformer 115 VAC to 12 VAC		
T5	Step down transformer 115 VAC to 3.0 VAC	M2	DC millimeter 0 - 300 mA
D1, D2	1N540	M3	Peak reading voltmeter or RMS equivalent (0 - 10 volts)
R1	20 kΩ, 1/2 watt		

FIGURE 6. Operational life test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Destructive physical analysis when requested.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil .

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6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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